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A Public-ID-Rejoinable DESI DR1 Anomaly-Candidate Catalog: 181 Warning-Free Primary Spectra from a Reproducible Positional Recovery

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ABSTRACT

Anomaly searches become reusable catalogs only when rows trace public archive objects and the selection function is reconstructable. We recover candidates from a historical DESI anomaly stream whose legacy identifiers mixed public-looking values with internal hashes. Starting from 190,015 positional clusters containing a DESI member, we stream all 28,425,963 rows of the public DESI Data Release 1 (DR1) pixel-based redshift catalog in 200,000-row chunks, read 18 columns, and query a spherical-coordinate k-d tree. The predeclared parent selection is a $1''$ match to a main-survey row carrying at least one LRG, ELG, QSO, BGS_ANY, or MWS_ANY `DESI_TARGET` bit. It reproduces exactly 20,299,155 eligible DESI rows and 2,468 positional matches. Requiring the global primary redshift row leaves 2,448; requiring `ZWARN`=0 leaves 181 unique public `TARGETID`s. These 181 rows form the v3.2.0 release. They comprise 157 Redrock GALAXY, 23 QSO, and one STAR classifications; spectral type was not a selection cut. Of the released matches, 170 are within $0.1''$ and 11 lie between $0.1''$ and $1''$; a quality-tier column exposes this tail without silently changing the declared match. Every released row and all 18 carried DESI fields were re-read from their recorded FITS row and compared exactly. The release includes the Parquet catalog, dictionary, code, waterfall, checksums, and provenance. This is a reproducible follow-up list, not validated detections or an unbiased sample for anomaly-rate inference.

Keywords: catalogs — surveys — methods: data analysis — methods: statistical — galaxies: spectra — quasars: general

1. INTRODUCTION

Spectroscopic surveys make it possible to search for unusual objects at scales that cannot be inspected manually. Reconstruction models, latent-variable models, and other outlier methods have consequently been applied to SDSS and DESI spectra [2, 3, 4]. Their scientific value, however, depends on a distinction that is easy to blur: an unusual model score is a candidate-selection signal, whereas an archive-resolvable object with auditable quality metadata is a usable catalog row. Neither is, by itself, a validated astrophysical detection.

This paper addresses that distinction through a structural recovery. An earlier multi-survey anomaly product contained DESI sky positions, anomaly scores, and residual summaries, but most legacy identifiers were internal hashes rather than public DESI `TARGETID`s. A catalog whose coordinates cannot be joined deterministically to public spectra is difficult to verify, extend, or cite object by object. The earlier product also mixed survey-specific selection functions. Treating its aggregate row count as a homogeneous scientific sample would therefore be misleading even if every row were numerically correct.

We replace that scope with one falsifiable deliverable: a DESI DR1 science-target candidate catalog whose public identifiers and carried metadata can be rejoined exactly. The recovery uses the historical anomaly stream only as a source of coordinates and score metadata. Public identity, targeting information, primary-row status, redshift metadata, and fit warnings all come from the current public DESI DR1 redshift catalog [1]. The result is a small quality-controlled follow-up set rather than a claim about the prevalence, novelty, or physical origin of anomalous spectra.

The contributions are:

1. a memory-bounded, checkpointed join of 190,015 DESI-containing anomaly clusters to the 28.4 million-row DR1 redshift catalog;

2. an exact reproduction of the previously quoted 2,468-row main-survey science-bit cohort;
3. a declared quality waterfall yielding 181 warning-free global-primary candidates;
4. a public-ID release with exact field-by-field source-row validation; and
5. a clean v3.2.0 bundle containing only this DESI product and the artifacts required to audit it.

Section 2 defines the two inputs. Section 3 specifies the join, selection, and duplicate rules. Section 4 describes the released cohort without interpreting it as a population measurement. Sections 5 and 6 make the selection biases and reproducibility checks explicit.

2. INPUTS AND SCOPE

2.1. *Historical anomaly-cluster substrate*

The starting cluster table contains 378,480 coordinate clusters produced by a prior positional deduplication. Exactly 190,015 have a survey-membership string containing `desi_dr1`; 190,005 are DESI-only clusters, eight also contain an SDSS member, and two also contain a LAMOST member. Those historical cross-survey labels are retained only as provenance columns. They do not enter the v3.2.0 selection and no non-DESI table is included in the new release.

The accompanying DESI anomaly table has 195,829 rows with an internal identifier, right ascension, declination, a scalar score S, the band producing the largest residual, and three residual summaries. The legacy identifier is not treated as a public key. It mixes positive public-looking values with internal hashed values; 169,611 of the 195,829 values are negative. Within the released 181-row cohort, ten legacy identifiers are negative. This behavior is preserved as audit history in `original_internal_tid`, while the recovered public `targetid` is the only archive join key recommended for users.

When a cluster contains more than one DESI anomaly member, its canonical historical member is the one with the largest original score; ties are resolved by the smaller original row number. The score and residual fields are carried forward verbatim as ranking metadata. They are not recomputed in this work, and they are not interpreted as calibrated probabilities, physical significances, or novelty measures.

2.2. *Public DESI DR1 redshift catalog*

The public identity substrate is `zall-pix-iron.fits` from DESI DR1. The local file has 28,425,963 rows and 22,371,272,640 bytes. Its SHA-256 is `2d95ad99361039b556c402b49e0e7c84df5f00106dc5731d44476a58b128b49b`, equal to the value in DESI's current official checksum list. The FITS table contains repeat observations, survey and program labels, targeting bitmasks, Redrock redshift products, and flags defining primary rows. We read only the fields listed in Table 1; no 21 GB in-memory table is formed.

3. RECOVERY AND SELECTION METHOD

3.1. *Memory-bounded positional join*

All cluster coordinates are converted from (α, δ) to three-dimensional unit vectors,

$$\boldsymbol{x} = (\cos\delta\cos\alpha, \cos\delta\sin\alpha, \sin\delta), \tag{1}$$

and indexed once with a `cKDTree`. The FITS table is then traversed in blocks of 200,000 rows using `fitsio`. Numeric arrays are converted from FITS big-endian storage to native byte order before Parquet serialization. Within each block we first apply the survey and target bit mask, convert the remaining coordinates to unit vectors, and query the nearest cluster. Chord distances are converted back to great-circle arcseconds. Only matches within the predeclared $1''$ radius are retained.

Each completed block is written atomically to a small Parquet part. A JSON checkpoint records the input signature, completed block starts, cumulative row counts, and the last completed row; a fsynced JSON-lines progress file records the same counts with timestamps. Restarting with the same signature skips completed parts. A changed FITS path, size, column set, radius, chunk size, or cluster count invalidates the checkpoint rather than silently mixing runs. The full scan took 97.1 s on the audit machine and retained only 2,468 match rows. Runtime is reported for reproducibility, not as a cross-platform benchmark.

3.2. *Science-target parent cohort*

The parent cohort uses the exact historical bitmask definition so that the earlier 2,468 count can be tested. A row is eligible when

$$\begin{aligned} &\texttt{SURVEY} = \texttt{main}, \\ &\texttt{DESI\_TARGET} \,\&\, \left(2^0 \,|\, 2^1 \,|\, 2^2 \,|\, 2^{60} \,|\, 2^{61}\right) \neq 0, \end{aligned} \tag{2}$$

where bits 0, 1, 2, 60, and 61 denote LRG, ELG, QSO, BGS_ANY, and MWS_ANY, respectively. These classes are not mutually exclusive. The streamed scan finds 20,299,155 eligible rows, exactly reproducing the historical denominator. Their nearest-neighbor join yields 2,468 rows within $1''$, exactly reproducing the historical

Table 1. DESI fields read from each streamed FITS chunk

Group	Fields	Role in this work
Identity/position	TARGETID, TARGET_RA, TARGET_DEC	public rejoin and angular match
Observing context	SURVEY, PROGRAM	main-survey selection and description
Targeting	DESI_TARGET, BGS_TARGET, MWS_TARGET, SCND_TARGET	science-bit selection and raw metadata
Redshift fit	Z, ZWARN, SPECTYPE, DELTACHI2	quality gate and descriptive metadata
Coadd status	COADD_FIBERSTATUS	descriptive quality cross-check
Primary flags	MAIN_NSPEC, MAIN_PRIMARY, ZCAT_NSPEC, ZCAT_PRIMARY	repeat accounting and final primary gate

NOTE—Column names follow the public DESI zcatalog. Only SURVEY, DESI_TARGET, ZCAT_PRIMARY, and ZWARN enter the final Boolean selection; coordinates enter the positional join. SPECTYPE and Z are not selection cuts.

match count. This agreement tests the coordinate transform, endian-correct bitmask evaluation, row traversal, and radius together.

3.3. *Quality cohort and duplicate rules*

The public redshift catalog may contain multiple rows associated with one target. We therefore require ZCAT_PRIMARY=true, the global primary redshift-row flag. We then require ZWARN=0. This latter gate is intentionally conservative: it removes any Redrock fit carrying a warning bit. We do not require a particular SPECTYPE, redshift range, or minimum DELTACHI2; imposing such post-hoc cuts would redefine which anomaly morphologies survive.

Before the final gates, a FITS row is assigned to its nearest anomaly cluster only. Within any cohort, duplicate cluster matches are ordered by smaller angular separation, larger DELTACHI2, smaller TARGETID, and smaller FITS row. A second pass uses the same ordering to enforce one row per public TARGETID. No tie-breaking is needed in the final cohort, but publishing the rule makes reruns deterministic. The final 181 rows have unique candidate IDs, cluster IDs, public TARGETIDs, and FITS row numbers.

3.4. *Match-quality tier*

The $1''$ radius was fixed before inspecting the quality cohort and is retained for exact historical count reproduction. Most recovered coordinates agree much more closely: 170 of 181 are separated by at most $0.1''$. Eleven lie between $0.1''$ and $1''$, including eight above $0.5''$ and five above $0.75''$; the maximum is $0.9906''$. Rather than silently deleting these valid-by-definition matches, we publish match_quality_tier with values coordinate_consistent_le_0p1arcsec and positional_match_gt_0p1_le_1arcsec. Users can reproduce either the declared catalog or the 170-row high-coordinate-consistency subset.

Table 2. Declared selection waterfall

Stage	Rows
DESI-containing anomaly clusters	190,015
Main science-bit DESI rows searched	20,299,155
$\leq 1''$ positional matches	2,468
Remove non-ZCAT_PRIMARY	-20
Global-primary matches	2,448
Remove primary rows with ZWARN$\neq 0$	$-2,267$
Released candidates	**181**

NOTE—The 181-row release is 7.33% of the positional science-bit cohort.

4. CATALOG RESULTS

4.1. *Selection waterfall*

Table 2 and Figure 1 give the full waterfall. Of the 2,468 parent matches, only 20 are removed by the global-primary requirement. The dominant reduction is ZWARN=0: 2,267 of the 2,448 primary rows carry at least one warning bit, leaving 181. Requiring MAIN_PRIMARY in addition changes nothing; requiring COADD_FIBERSTATUS=0 after the adopted gates also changes nothing. These equalities are reported as cross-checks, not as independent evidence that the sample is unbiased.

The large ZWARN reduction is scientifically important. An anomaly-selected spectrum is more likely than an ordinary spectrum to be difficult for a standard template-fitting pipeline. Consequently, the warning-free subset is cleaner for catalog-level redshift metadata but is not a random subset of all unusual spectra. This tradeoff motivates the restricted interpretation in Section 5.

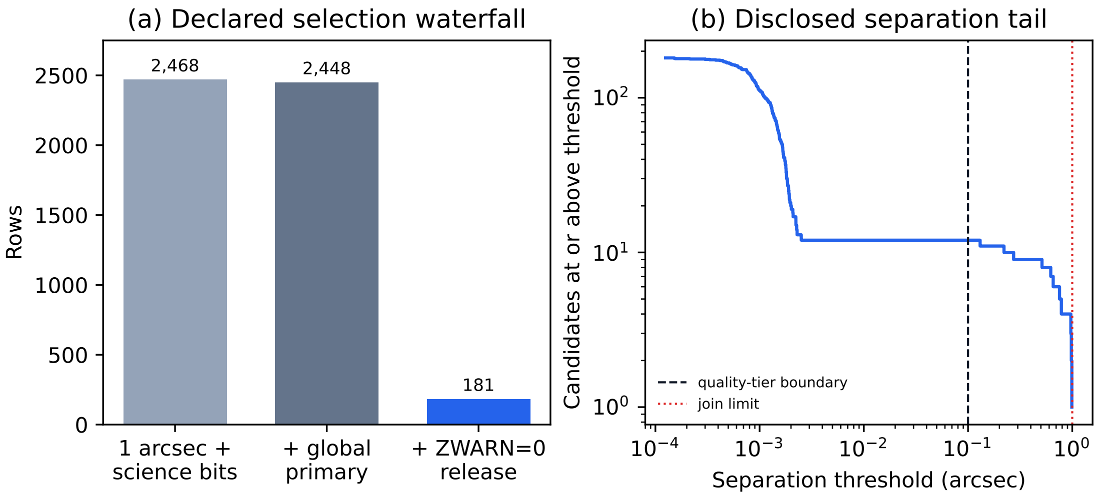

Figure 1. *Left:* Selection waterfall from the exact historical bitmask cohort to the warning-free global-primary release. *Right:* Number of released candidates at or above each positional separation. The dashed line marks the published quality-tier boundary; the dotted line is the predeclared join limit.

4.2. *Descriptive composition*

Redrock classifies 157 released rows as GALAXY, 23 as QSO, and one as STAR. The observing programs are 162 dark and 19 bright. Target-bit labels are overlapping: 157 rows carry ELG, 18 BGS_ANY, seven QSO, one LRG, and one MWS_ANY; a row can contribute to more than one count. These values describe the catalog after selection. They do not measure selection efficiencies by class because the relevant class-specific parent denominators and warning mechanisms have not been modeled.

The redshift metadata span $-0.00034 \leq z \leq 6.0878$, with quartiles 0.616, 1.108, and 1.401. Two rows have $z < 0$; 36 lie in $0 \leq z < 0.5$, 36 in $0.5 \leq z < 1$, 76 in $1 \leq z < 1.5$, eight in $1.5 \leq z < 2$, 13 in $2 \leq z < 3$, and ten at $z \geq 3$. The two small negative values are retained because `ZWARN`=0 is the declared quality criterion and no physical-redshift cut was predeclared. Users interested in a positive-redshift science subset should add that cut explicitly and report the resulting 179 rows.

The original anomaly-score range in the release is 5.0006–7.7057, with a median of 5.3244. This narrow range is inherited from the historical anomaly threshold. We do not convert it to a Gaussian significance or compare it across models. Figure 2 shows the sky, redshift, score, and separation metadata without attaching a detection interpretation.

4.3. *Sky and positional coverage*

The released positions range from 3.73° to 352.15° in right ascension and −19.31° to 75.49° in declination. There are 134 candidates north and 47 south of the celestial equator. All twelve 30-degree right-ascension bins are occupied, while five of six coarse equal-area declination bins are occupied. These summaries demonstrate geographic coverage within the input, not isotropy. The footprint is inherited from DESI DR1 and the historical anomaly sample; no all-sky uniformity correction is attempted.

The median positional separation is $0.00127''$, with quartiles $0.00083''$ and $0.00167''$. The abrupt tail of 11 larger separations is why both the continuous separation and categorical quality tier are released. Positional coincidence inside $1''$ supports a catalog association, but it does not prove that the historical anomaly-row identity and public spectrum are physically identical in crowded or blended fields. Those 11 rows are natural priorities for image-level and spectrum-level manual verification.

4.4. *Example rows*

Table 3 lists the twelve highest historical scores only to illustrate the schema. Rank in this table is not a recommendation that the objects are novel or astrophysically important. The complete machine-readable catalog contains all 181 rows.

5. SELECTION FUNCTION AND APPROPRIATE USE

5.1. *What the catalog selects*

The release is the intersection of four operations: membership in a historical anomaly-cluster substrate, positional coincidence with a public DESI row inside $1''$, membership in the declared main-survey science-bit parent, and the `ZCAT_PRIMARY`/`ZWARN`=0 quality gates. It is therefore conditional on the historical anomaly model, preprocessing, score threshold, survey footprint,

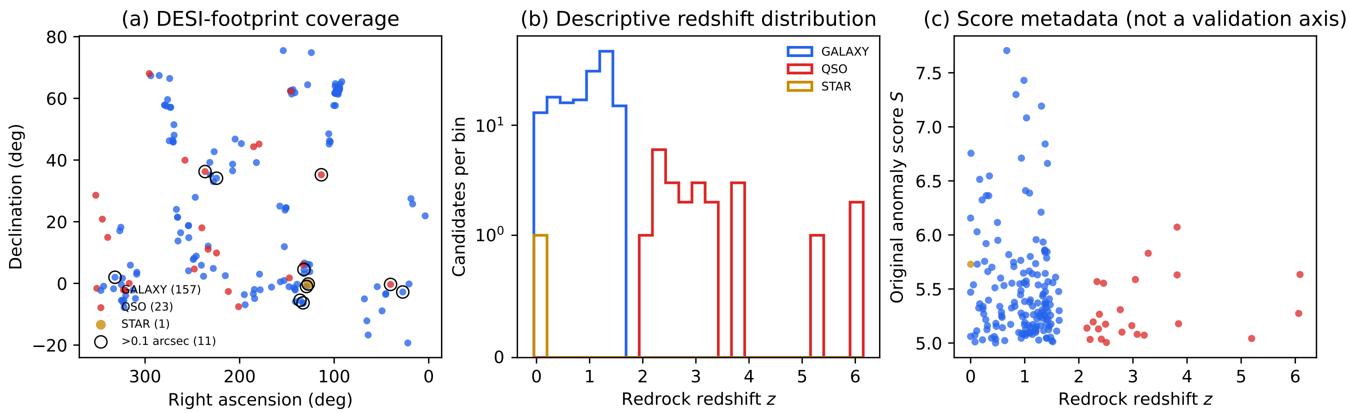

Figure 2. Released v3.2.0 candidates. (a) DESI-footprint sky distribution colored by descriptive Redrock `SPECTYPE`; open circles mark the 11 rows in the 0.1–1″ separation tier. (b) Redshift metadata by spectral type. (c) Historical anomaly score versus Redrock redshift. Neither `SPECTYPE` nor redshift was used to select the release, and the score is not a calibrated physical significance.

Table 3. Highest-score example rows from the released catalog

Candidate	TARGETID	R.A.	Decl.	Type	z	S	Sep. (arcsec)
P3-DESI-000001	39633349869831142	272.82236	57.12256	GALAXY	0.668	7.71	0.001
P3-DESI-000002	39627706974869534	45.28046	-3.31238	GALAXY	0.986	7.43	0.002
P3-DESI-000003	39627895609492977	131.94836	4.51377	GALAXY	0.838	7.30	0.274
P3-DESI-000004	39633407088529181	97.49930	61.65664	GALAXY	1.310	7.19	0.001
P3-DESI-000005	39627636166626827	132.90847	-6.30883	GALAXY	1.033	7.08	0.991
P3-DESI-000006	39633418404759790	95.13019	62.74366	GALAXY	1.376	6.84	0.000
P3-DESI-000007	39636182677589446	64.05184	-16.74091	GALAXY	0.006	6.76	0.002
P3-DESI-000008	39633407096917948	98.59444	61.74095	GALAXY	0.938	6.71	0.000
P3-DESI-000009	39627748041295714	333.31949	-1.73961	GALAXY	1.420	6.66	0.001
P3-DESI-000010	39633186518469218	104.94924	46.24594	GALAXY	0.347	6.55	0.001
P3-DESI-000011	39633063570837088	231.50666	39.22886	GALAXY	0.167	6.51	0.001
P3-DESI-000012	39627877414602764	125.74951	3.78702	GALAXY	1.015	6.41	0.002

Note—R.A. and declination are ICRS degrees. Type and z are public Redrock metadata. S is preserved historical anomaly-score metadata. Values are rounded for display only.

target selection, public zcatalog construction, and Redrock warning behavior. The present work makes those operations reproducible; it does not make them ignorable.

In particular, the released fraction $181/2468 = 7.33\%$ must not be called an anomaly rate, a validation efficiency, or a purity estimate. The numerator is conditioned on warning-free template fits, while the denominator contains the very spectra for which template fitting may be difficult. A population analysis would require at least the class-, magnitude-, redshift-, signal-to-noise-, and warning-dependent inclusion probabilities for both the historical anomaly stage and the DESI re-join stage. Those quantities are not available in this release.

5.2. *Recommended uses*

The catalog is designed for:

- deterministic retrieval of 181 public DESI spectra by `TARGETID`;
- manual or automated spectral vetting using the full public data products;
- testing alternative anomaly models on a fixed, warning-free candidate list;

- follow-up prioritization using continuous score, separation, target-bit, and redshift metadata; and
- reproducibility exercises in archive recovery and catalog selection.

It is not designed for estimating an anomaly occurrence rate, claiming new object classes, measuring cosmological parameters, or comparing raw counts with a differently selected survey. No SIMBAD-based novelty fraction, process-volume multiplier, cross-survey total, or cosmological inference is part of v3.2.0. Establishing novelty requires object-by-object literature and archive review beyond a coordinate database match. Establishing a physical anomaly requires inspection of calibrated spectra, masks, observing conditions, and repeat observations.

5.3. *Relationship to prior DESI anomaly searches*

Prior DESI studies have demonstrated useful outlier-search strategies for particular target classes and data subsets [3, 4]. Direct candidate-count comparisons are not meaningful here because the parent populations, preprocessing, models, thresholds, and quality gates differ. The contribution of v3.2.0 is orthogonal: it converts a historically unrejoinable coordinate-and-score product into a small public-key catalog with a fully enumerated waterfall. Future model comparisons can use these 181 public spectra as a common follow-up set, but evaluation against a representative control sample would still be required.

6. REPRODUCIBILITY AND INTEGRITY AUDIT

6.1. *Exact source-row rejoin*

The independent validator reads the released `fits_row` values from the final Parquet, reopens the public FITS file, and retrieves all 18 source fields listed in Table 1. Integer, Boolean, and string fields are compared exactly; floating fields are also compared at zero tolerance because the Parquet values were copied directly. All 181 rows pass all 18 field comparisons. The validator separately confirms unique candidate, cluster, `TARGETID`, and FITS-row keys; zero null cells; main-survey membership; at least one declared science bit; `ZCAT_PRIMARY`=true; `ZWARN`=0; and separation at or below $1''$.

6.2. *Provenance and checksum reconciliation*

The full local 22.37 GB FITS SHA-256 was recomputed. A no-cache retrieval of DESI's current official checksum file reports the same value. The live FITS endpoint reports the same byte size, an ETag of `6a516695-5356e87c0`, and a last-modified time of 2026-07-10 21:39:33 GMT. Eight independently chosen 1 MiB byte ranges, including the first and last ranges, also match between the local file and live endpoint. An older local provenance sidecar written in May 2026 records a different pre-replacement SHA-256. We preserve that stale value in `PROVENANCE.json` and explicitly exclude it from current validation rather than overwriting the history.

The two historical input Parquets are also hashed: the cluster table is `b14deb02...d6138c643` (12,087,824 bytes) and the anomaly table is `0a36b8d6...f103ec65` (10,514,503 bytes). Full hashes, not abbreviated display forms, are in the release manifest and provenance file.

6.3. *Release contents*

The clean v3.2.0 directory contains one science table and its audit artifacts:

1. desi_dr1_science_anomaly_candidates_v3.2.0.parquet;
2. `DATA_DICTIONARY.md` and `README.md`;
3. `COHORT_COUNTS.json`, `QC_REPORT.json`, and `SELECTION_AUDIT.json/.md`;
4. exact build and independent-validation scripts;
5. `PROVENANCE.json`; and
6. `RELEASE_MANIFEST.json` with SHA-256 and byte size for every payload file.

The manifest excludes itself to avoid a self-referential hash. An independent post-generation pass recomputes every listed hash and byte count. The release contains no Gaia, eROSITA, LAMOST, SDSS, Planck, ACT, or heterogeneous aggregate table. Earlier releases remain historical and are not moved or rewritten.

7. LIMITATIONS

The catalog has six central limitations.

First, historical score lineage. The original score and residual columns are preserved from the anomaly stream rather than recomputed from public spectra. Public identifiers now make future rescoring possible, but v3.2.0 does not claim exact reproduction of the historical neural network output.

Second, quality-conditioned incompleteness. Requiring `ZWARN`=0 removes 92.6% of primary positional matches. The released list is cleaner for metadata use but incomplete by construction and likely biased against the most difficult-to-fit spectra.

Third, positional association. A 1″ match is not a proof of identity in every field. The 11 rows above 0.1″ are flagged, and all users retain the continuous separation.

Fourth, no astrophysical vetting. Redrock class, redshift, and `DELTACHI2` are catalog metadata. We have not inspected every spectrum, image, mask, or repeat observation, and we make no novelty or physical-anomaly claim.

Fifth, inherited footprint and targeting. The sample follows DESI DR1 coverage and the declared target bits. Target classes overlap, and no completeness correction is provided.

Sixth, version specificity. Public redshift products can be regenerated or superseded. The release pins file size and checksum so later catalog versions can be compared rather than silently substituted.

These limitations are not peripheral caveats: they define the valid use of the product. The release should be cited as a candidate catalog and reproducibility artifact.

8. CONCLUSIONS

We have replaced a heterogeneous, public-ID-limited anomaly product with a focused DESI DR1 catalog whose selection and provenance are testable from end to end. A memory-bounded scan of 28,425,963 public zcatalog rows exactly reproduces the prior 20,299,155-row science-bit denominator and 2,468 one-arcsecond positional matches. The declared global-primary and warning-free gates yield 181 unique public `TARGETID`s. All carried DESI fields rejoin exactly, all released keys are unique, and every payload hash and byte size passes an independent manifest audit.

The smaller, auditable count is the catalog-integrity improvement. It replaces an unsupported implication of hundreds of thousands of validated detections with 181 traceable candidates and a clear account of the 2,287 exclusions. The release retains the full 1″ cohort definition while exposing the 11-row separation tail, preserves historical score metadata without turning it into a significance, and states that the warning-free cohort is unsuitable for occurrence-rate inference. Its immediate purpose is reproducible spectral retrieval and follow-up. Physical classification is future work.

DATA AVAILABILITY

The v3.2.0 release bundle is published under the immutable correction tag p3-v3.2.0-r1 in the public Hugging Face dataset https://huggingface.co/datasets/bamfai/bigbounce-anomaly-catalog, path `releases/p3-v3.2.0/`. The tag resolves to commit `983209ae606be311d9bda9f0258716d56386ee69`. All 11 remote files were downloaded from that tag and verified byte-for-byte against the local release. The original `p3-v3.2.0` tag remains immutable at commit `f2df8c00284350a7fe84a4ba6eb06468408b1c49`; p3-v3.2.0-r1 supersedes it only because the correction adds explicit immutable-input URLs and reproduction commands.

The historical cluster and anomaly inputs are respectively https://huggingface.co/datasets/bamfai/bigbounce-anomaly-catalog/resolve/p3-v3.1.161/pathc_unique_objects.parquet and https://huggingface.co/datasets/bamfai/bigbounce-anomaly-catalog/resolve/p3-v3.1.161/desi_dr1_anomalies.parquet. Both are pinned by `p3-v3.1.161`, which resolves to commit `cdaaa03a72c69d86f011be128d93f261dc5b39a8`; their SHA-256 values are recorded in `PROVENANCE.json`. The public DESI input and official checksum list are available at https://data.desi.lbl.gov/public/dr1/spectro/redux/iron/zcatalog/v1/. After downloading the three named inputs, the concrete build command is:

```
python3 build_desi_science_catalog_v320.py \
  --clusters pathc_unique_objects.parquet \
  --anomalies desi_dr1_anomalies.parquet \
  --fits zall-pix-iron.fits \
  --output-dir desi_science_catalog_v3.2.0 \
  --chunk-rows 200000 \
  --fits-sha256 \
  2d95ad99361039b556c402b49e0e7c84\
df5f00106dc5731d44476a58b128b49b
```

The shell joins the two physical hash lines because the first ends in a backslash. The release README gives the same copy-paste download, build, and validation commands. The source tree contains the same scripts at pipelines/p3_anomaly_engine/scripts/. No secret credentials or private data are required to validate the released catalog against the public DESI file.

SOFTWARE

The build used Python 3.9.6, NumPy 1.26.4, pandas 2.1.4, SciPy 1.13.1 [5], fitsio 1.3.0, and PyArrow 21.0.0. FITS is described by the FITS standard [6]. Exact versions and the chunk size are recorded in `PROVENANCE.json`.

ACKNOWLEDGMENTS

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manuscript review; all catalog assertions reported here were checked against machine-readable artifacts, and no review-system output was accepted as evidence without a local or public-source verification.

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APPENDIX

A. EXECUTABLE SELECTION SPECIFICATION

For clarity, the released cohort can be represented by the following ordered specification:

1. Select the 190,015 historical clusters with `desi_dr1` in their survey membership.
2. For each cluster, select the highest-score DESI historical member, breaking ties by source row.
3. Stream public zcatalog rows and retain `SURVEY==main` with at least one declared `DESI_TARGET` bit.
4. Assign each eligible zcatalog row to its nearest cluster in spherical unit-vector space.
5. Retain separations $\leq 1''$.
6. Within a cohort, keep the nearest row per cluster; break ties by descending `DELTACHI2`, ascending `TARGETID`, and ascending FITS row.
7. Apply the same ordering to keep one cluster per `TARGETID`.
8. Require `ZCAT_PRIMARY`=true and `ZWARN`=0.
9. Label separation $\leq 0.1''$ as coordinate-consistent and the remaining declared $1''$ matches as the positional tail.
10. Sort by `cluster_id` and assign sequential release-local candidate IDs.

The script writes checkpoint parts before forming the final cohort. Thus the selection counts can be inspected independently of the manuscript and an interrupted full-catalog scan can resume without replaying completed chunks.

B. COLUMN GROUPS

The release contains 42 columns after addition of the separation-quality tier. The authoritative field-by-field descriptions and storage dtypes are in `DATA_DICTIONARY.md`. The two tables below define the public and historical field groups without duplicating them in a third summary table.

Table 4. Release and public-DESI field definitions

Column	Definition
`candidate_id`	Stable release-local ID ordered by cluster ID.
`fits_row`	Zero-based row in the public DR1 ZCATALOG extension used by the exact rejoin audit.
`cluster_table_row`	Zero-based row in the committed historical cluster table.
`match_separation_arcsec`	Great-circle target-to-cluster separation in arcseconds.
`match_quality_tier`	At most $0.1''$, or greater than $0.1''$ and at most $1''$.
`targetid`	Public DESI `TARGETID`; recommended archive key.
`target_ra`, `target_dec`	Public DESI target coordinates in ICRS degrees.
`survey`, `program`	DESI survey and observing-program labels.
`desi_target`	Raw `DESI_TARGET` bitmask used for the declared science-bit gate.
`bgs_target`, `mws_target`, `scnd_target`	Raw public class-specific targeting bitmasks.
`z`	Redrock redshift estimate; descriptive, not a selection label.
`zwarn`	Redrock warning mask; required to equal zero.
`spectype`	Redrock best-fit spectral type; not used as a selection cut.
`deltachi2`	Best-versus-next-best Redrock template $\Delta\chi^2$.
`coadd_fiberstatus`	Bitwise OR of fiber-status flags contributing to the coadd.
`main_nspec`	Number of associated main-survey spectra.
`main_primary`	Primary-within-main-survey flag.
`zcat_nspec`	Number of spectra in the global zcatalog grouping.
`zcat_primary`	Global primary redshift-row flag; required to be true.

Note—Raw integer bitmasks are preserved so future DESI software can decode more detailed classes without relying on the release's compact label.

Table 5. Historical-cluster and anomaly-metadata field definitions

Column	Definition
`cluster_id`	Stable identifier from the historical positional clustering.
`n_detections`	Number of historical anomaly members in the cluster.
`n_surveys`	Number of historical survey labels in the cluster.
`survey_list`	Comma-separated historical membership; audit provenance only.
`cluster_ra_deg`, `cluster_dec_deg`	Mean historical cluster coordinate in ICRS degrees.
`cluster_best_score`	Maximum historical score over all cluster members.
`member_ids`	Pipe-separated legacy row labels; not public archive keys.
`best_survey`	Historical survey supplying the cluster-best score.
`desi_source_row`	Row of the canonical DESI member in the historical anomaly table.
`original_internal_tid`	Legacy mixed/hash identifier; negative values are expected and it must not be used as a public key.
`original_ra_deg`, `original_dec_deg`	Coordinates of the canonical historical DESI anomaly member.
`original_score`	Preserved historical anomaly-ranking score, not a calibrated significance.
`original_worst_band`	Band with the largest historical residual summary.
`original_residual_b`	Preserved historical B-band residual summary.
`original_residual_r`	Preserved historical R-band residual summary.
`original_residual_z`	Preserved historical Z-band residual summary.
`science_target_class`	Pipe-separated decoded selected bits; labels are nonexclusive.

Note—These fields preserve the recovery lineage. They do not convert historical model output into public identity or astrophysical validation.

C. AUDIT MATRIX

Table 6. Independent v3.2.0 integrity checks

Check	Result	Evidence
Historical denominator	PASS: 20,299,155	streamed cumulative counter
Historical match count	PASS: 2,468	checkpoint parts and cohort JSON
Strict cohort count	PASS: 181	independent recomputation from parts
Public identity	PASS: 181/181	direct FITS-row `TARGETID` comparison
Carried DESI fields	PASS: 18/18 fields for 181 rows	zero-tolerance source-row comparison
Uniqueness	PASS	candidate, cluster, TARGETID, and FITS row
Null cells	PASS: none	full Parquet null scan
Selection gates	PASS	main, science bit, primary, warning-free, radius
FITS provenance	PASS	full local SHA equals current official checksum
Remote parity	PASS	size plus eight sampled 1 MiB ranges
Payload manifest	PASS: 10/10 files	independent SHA-256 and byte-size recomputation

Note—PASS describes integrity relative to the declared release contract. It is not a claim that the candidates are astrophysically anomalous or novel.